PRACTICAL - 4

Aim:

1. Password Strength Checker Aim:

To develop a Java program that checks password strength by validating security rules such as minimum length, uppercase letters, digits, and special characters, and classifies the password as Weak, Medium, or Strong.

Code Screenshot :

prac 4 > PasswardChecker.java

```
1   import java.util.Scanner;
2
3   public class PasswardChecker {
4
        Run | Debug
5       public static void main(String[] args) {
6           int c = 0;
7           Scanner input = new Scanner(System.in);
8           String pass;
9
10          do {
11              System.out.println(x: "Password must have at least 8 characters.");
12              System.out.println(x: "Enter the password:");
13              pass = input.nextLine();
14          } while (pass.length() < 8);
15
16
17          boolean u = pass.chars().anyMatch(Character::isUpperCase);
18          if (u) c++;
19          boolean l = pass.chars().anyMatch(Character::isLowerCase);
20          if (l) c++;
21           boolean d = pass.chars().anyMatch(Character::isDigit);
```

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```java
prac 4 > J PasswardChecker.java
  3   public class PasswardChecker {
  5       public static void main(String[] args) {
18            if (u) c++;
19            boolean l = pass.chars().anyMatch(Character::isLowerCase);
20            if (l) c++;
21            boolean d = pass.chars().anyMatch(Character::isDigit);
22            if (d) c++;
23            boolean sc = pass.chars().anyMatch(ch -> !Character.isLetterOrDigit(ch));
24            if (sc) c++;
25
26            if (c <= 1)  System.out.println(x: "weak");
27            if (c == 2 || c == 3)  System.out.println(x: "medium");
28            if (c == 4) System.out.println(x: "Strong");
29
30            input.close();
31        }
32    }
33
```

OUTPUT :

```
sers\Administrator\AppData\Roaming\Code\User\worksp
Password must have at least 8 characters.
Password must have at least 8 characters.
Enter the password:
Enter the password:
M@isti#9080
Strong
```

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```java
import java.util.Scanner;

public class pr_4_driver {

    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println(x: "Enter passwords to check (type 'quit' to stop)");

        while (true) {
            System.out.print(s: "Password: ");
            String pw = sc.nextLine();

            if (pw.equalsIgnoreCase(anotherString: "quit")) {
                break;
            }

            String result = passwordchecker.strength(pw);
            System.out.println("Result: " + result);
        }

        sc.close();
    }
}
```

Output :

```
Enter passwords to check (type 'quit' to stop)
Password: abcdefgh
Result: Weak
Result: Weak
Password: 1234fghi
Password: 1234fghi
Result: Medium
Password: M@isti9080
Result: Strong
Password:
```

CEUE203 – OOP 25IT028 **Learning Outcome 1:**

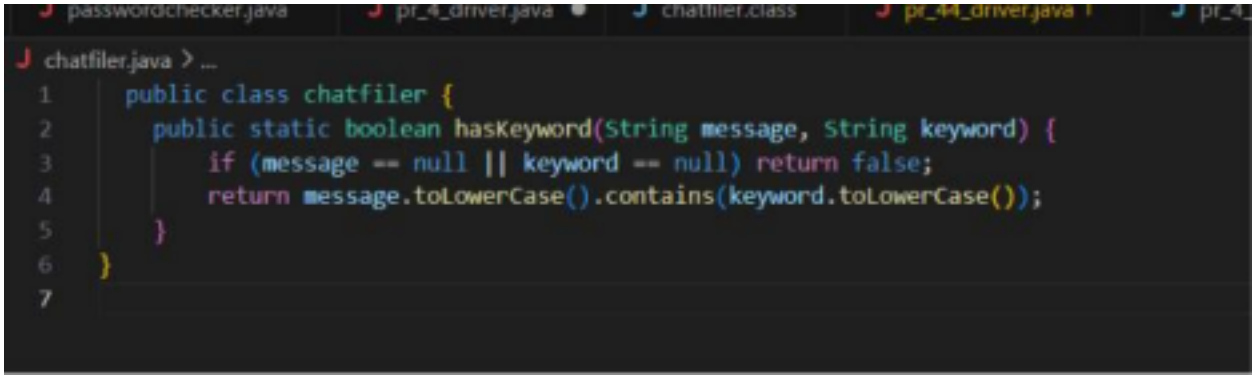
I have learned how to use String methods and regular expressions in Java. I have understood password validation techniques and method implementation. I have also improved my logical thinking and problem-solving skills through condition checking.

2. Chat-Log Filter

Aim:

To develop a Java program that filters chat log messages based on a user-provided keyword, ignores malformed log entries, and displays all matching messages along with the total number of matches.

Code Screenshot 2:



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```java
44_driver.java > pr_44_driver
import java.util.Scanner;

public class pr_44_driver {
    Run | Debug
    public static void main(String[] args) {
        String[] logs = {
            "10:05 alice Hello there",
            "10:10 bob How are you?",
            "10:15 carol hello everyone"
        };

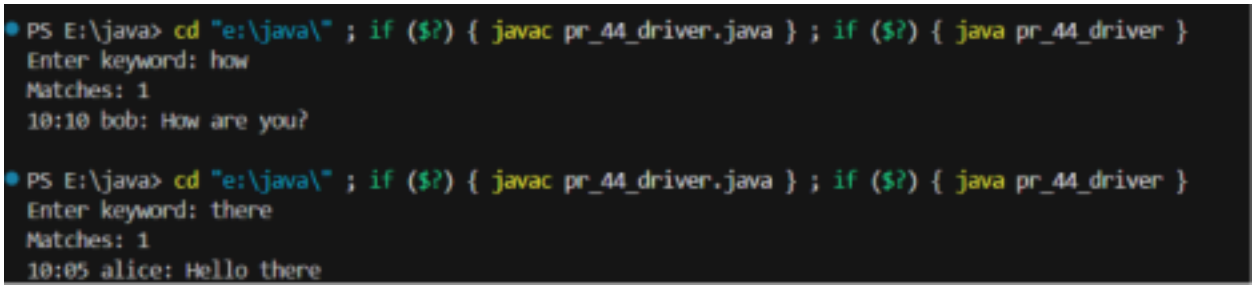
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter keyword: ");
        String keyword = sc.nextLine();

        int count = 0;
        StringBuilder report = new StringBuilder();

        for (String line : logs) {
            String[] parts = line.split(regex: " ", limit: 3);
            if (parts.length < 3) continue;
            String time = parts[0];
            String user = parts[1];
            String msg = parts[2];
            if (chatfiler.hasKeyword(msg, keyword)) {
                count++;
                report.append(time).append(str: " ").append(use
                        .append(str: ": ").append(msg).append(str
            }
        }

        System.out.println("Matches: " + count);
        System.out.println(report.toString());
    }
}
```

Output Screenshot 2:



Learning Outcome 2:

I have learned how to process and manipulate strings in Java. I have understood the use of split(), StringBuilder, and case-insensitive searching. I have also gained experience in handling invalid input and generating formatted output.